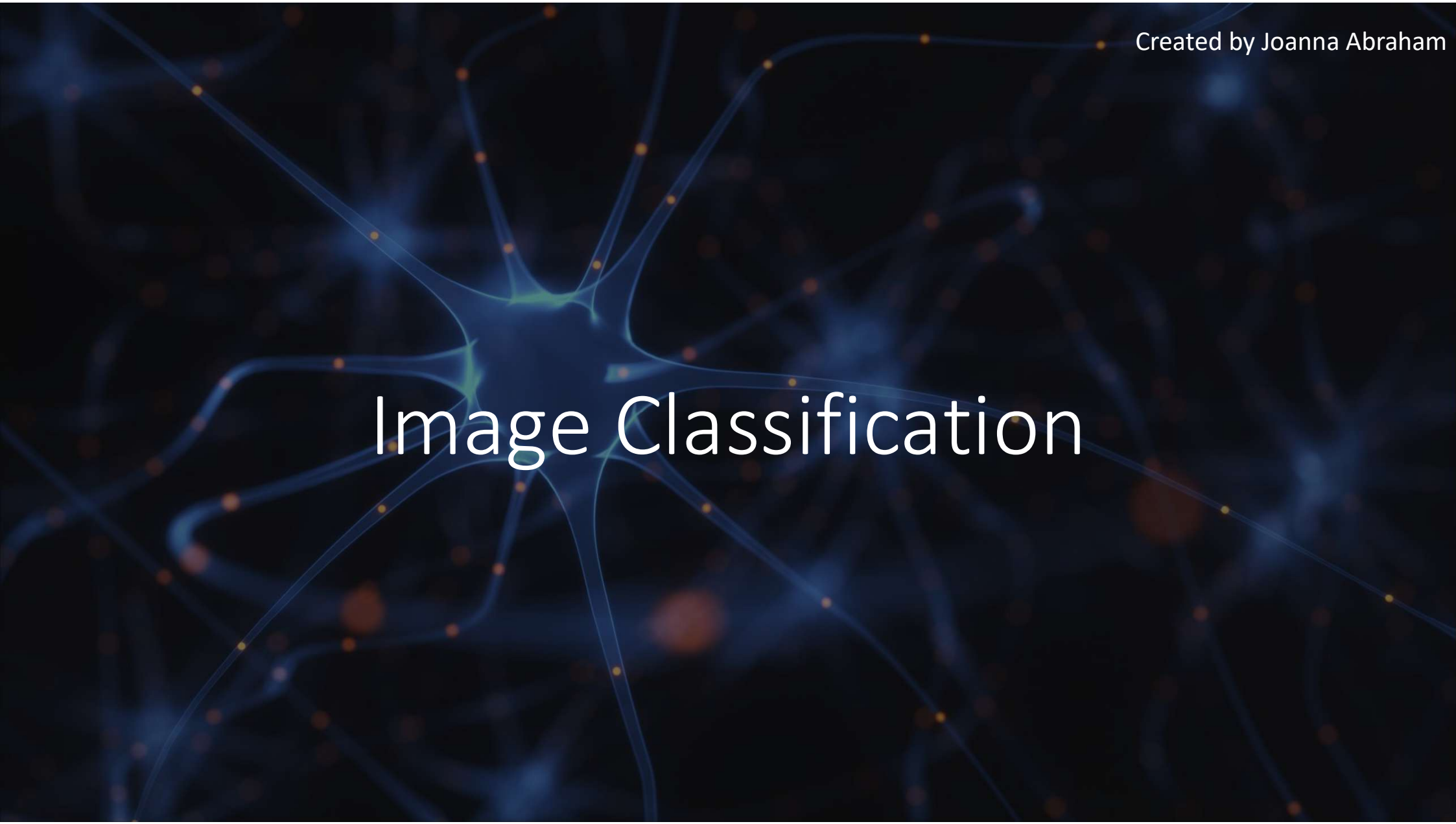
Automation of Wheel Retrieval, Deburring, Polishing and Storage

Group 12

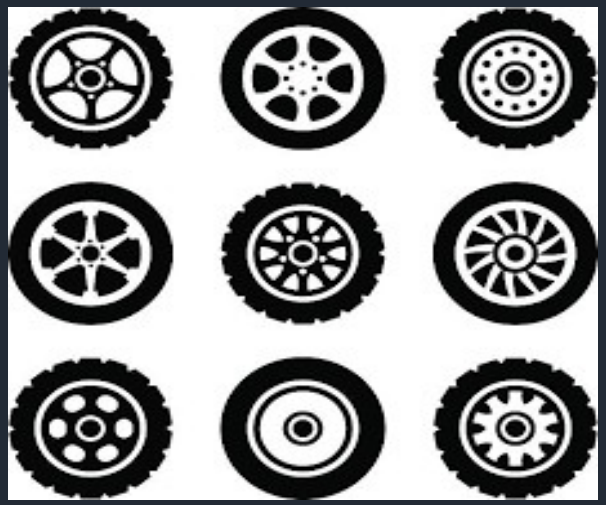


Created by Joanna Abraham

Image Classification

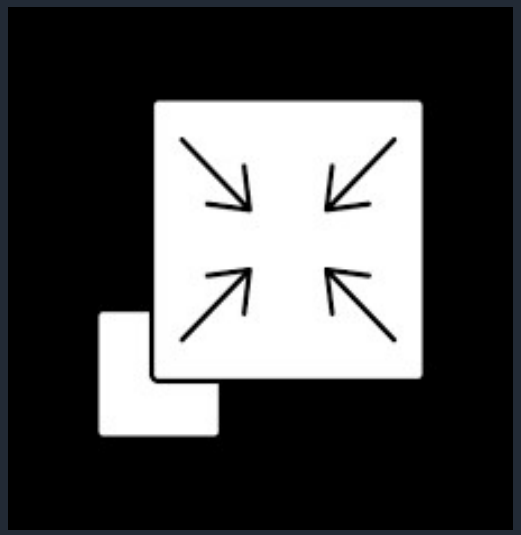
Dataset Collection

- 10 rim types with 15 images each
- Split into:
  - 70% training
  - 20% validation
  - 10% testing



Data Pre-processing

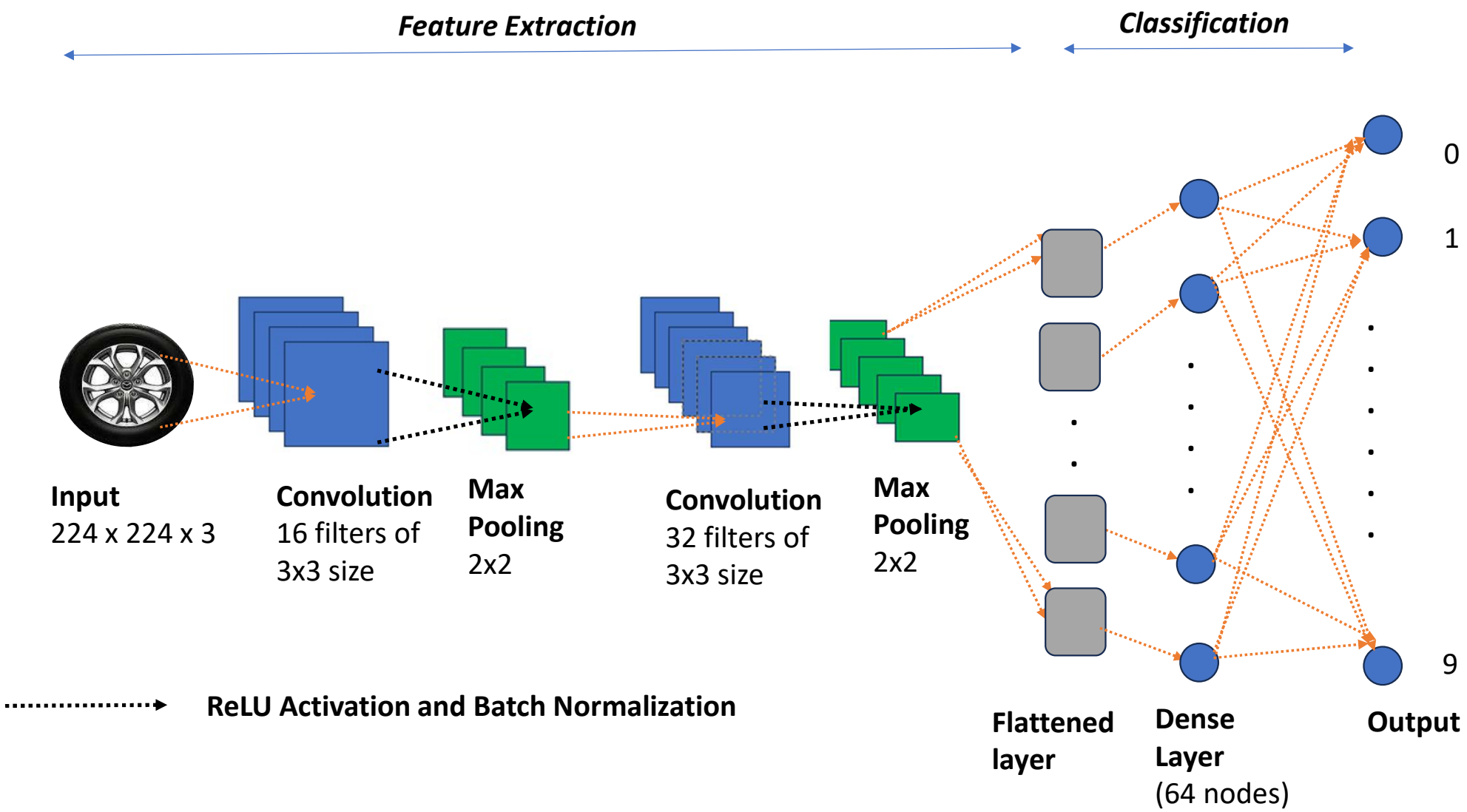
- Resize images to 224 x 224 pixels
- Crop or pad images to ensure consistent input size
- Normalize pixel values to the range [0, 1]



Created by Joanna Abraham

Convolutional Neural Network

Created by Joanna Abraham



Created by Joanna Abraham

Evaluation Techniques

Created by Joanna Abraham

1. Performance Metrics

- Accuracy: Overall correct classification rate
- Precision: Proportion of correct positive predictions per class
- Recall: Proportion of actual positives correctly identified
- F1-Score: Harmonic mean of precision and recall

2. Confusion Matrix Analysis

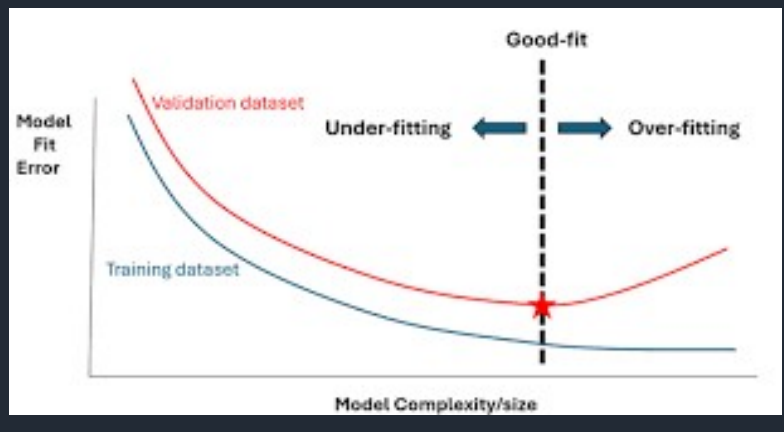
- Visualize misclassification patterns and identify:
  - Most frequently confused rim types
  - Classes with the highest/lowest performance
  - Potential similarities between misclassified classes

Confusion Matrix

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

3. Overfitting Detection

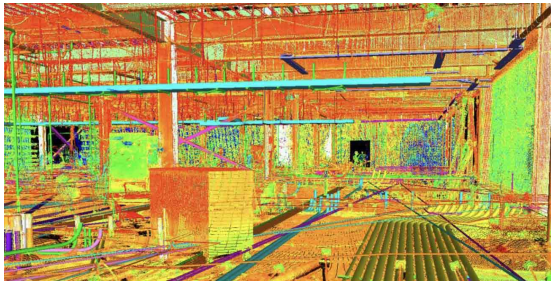
- Compare training vs validation:
  - Accuracy curves
  - Loss curves
  - Learning rate tracking



Rim Retrieval from Disorganized Storage

Drone Scanning

- Grid-based scanning
- LiDAR → **3D depth map**



LiDAR Environment Scan

Image Processing

- Embedded CNN → Rim Classification
- **Optimal Rim Retrieval Algorithm**

① Shortest travel distance
② Surface accessibility
③ Rim clustering
④ CNN classification confidence

Data Relaying

- Communication between stations
- **Rim data preloading**

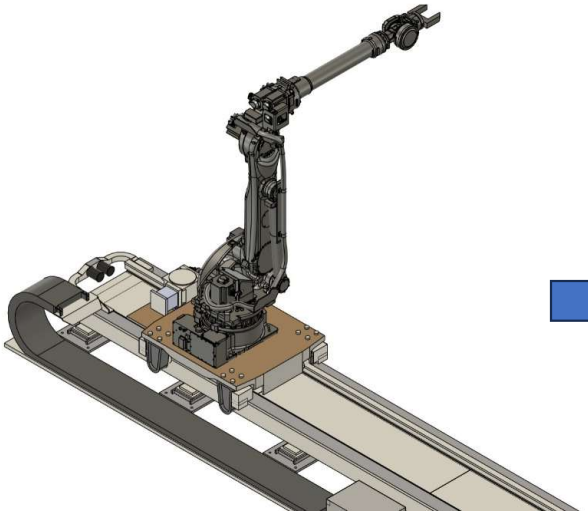
Robotic Arm Pickup

- External mount on linear axis → **Extended operational workspace**
- Dynamic gripping method based on **orientation**

Slide by Rahul Jo-Shua

Robotic Arm Pickup Models

Initial Vention Design



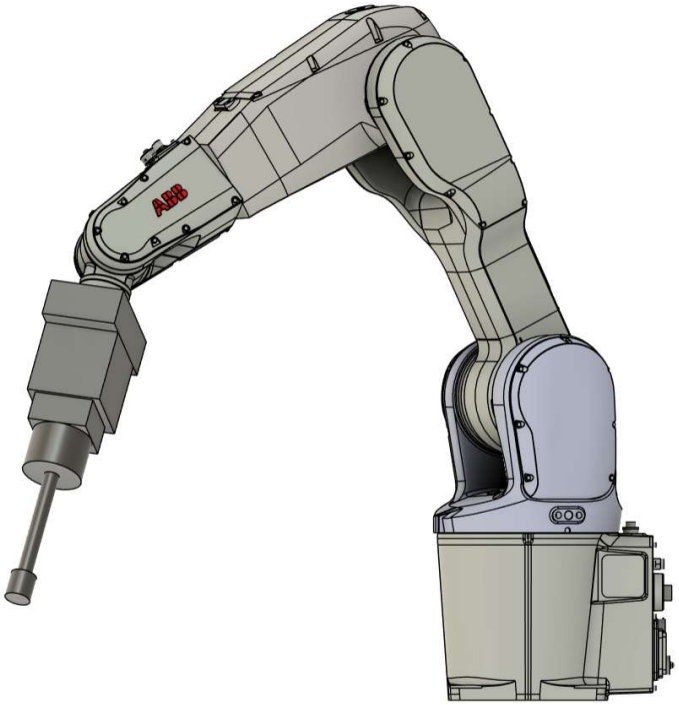
Developed CAD Design

Robotic Arm Simulation on Linear Axis

Slide by Rahul Jo-Shua

Deburring Station

- 6-DoF Robotic Arm
- Force-Torque Sensor at End-Effector
- Preloaded Rim-Specific Parameters
- Debris Management System → Vacuum Integrated



Deburring Arm CAD Model

Slide by Rahul Jo-Shua

Flipping Mechanism

Gravity-Assisted Flipping

Rim flips over naturally once its center of gravity passes the support point — no actuators needed.

Optimized Curve Design – Linear angular increase of $55^o/m$

Gradual incline ensures rims of varying sizes safely tilt and flip within a compact 2m length.

Damage Prevention and Safe Landing

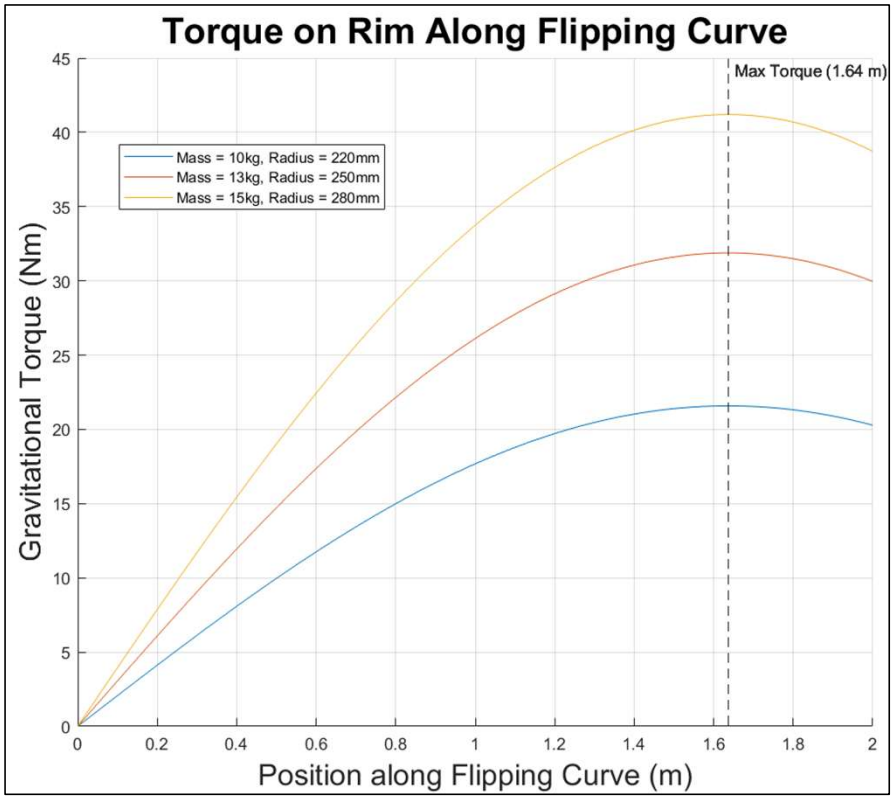
Curved descent and wider gap absorb impact, avoiding rim bounce or upright landing.

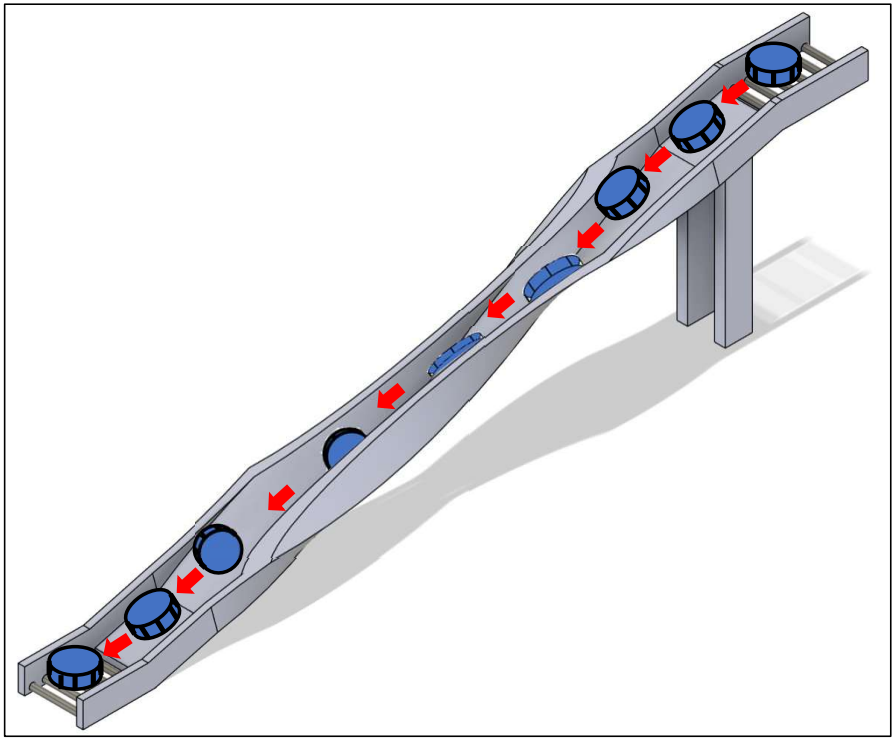
Post-Flip Re-centering

Funnel-shaped conveyor section re-aligns rim to the center for accurate polishing and deburring.

Nikolaos Mitsokapas

Final Design

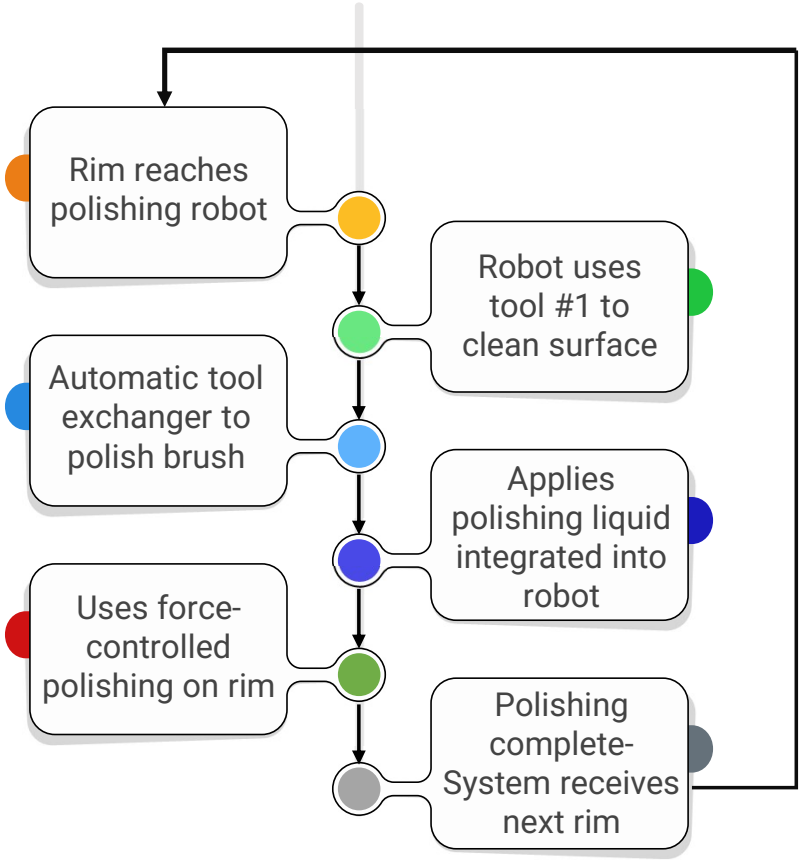


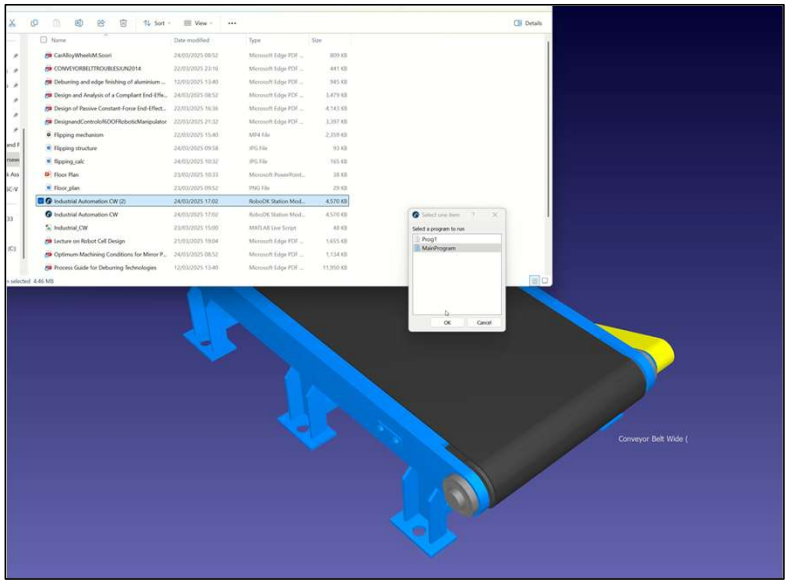


Nikolaos Mitsokapas

Polishing Station

- 6 DoF Robotic System with tool exchanger
- Surface Cleaning for Quality Prep
- Polishing Fluid Application
- Force-Controlled Precision Polishing





Nikolaos Mitsokapas

PART.4

storage system

An intelligent single-unit storage system for unwrapped wheel hubs, designed to support flexible order handling, space efficiency, and full automation without pallets.

Combining passive positioning with AMR-based retrieval, it ensures surface protection and scalable storage capacity.

Design Motivation and Assumptions

No packaging

Unwrapped Output

The rim exit the production line unprotected, requiring immediate surface-safe handling.

Model Variability

Different rim necessitate per-unit identification and storage.

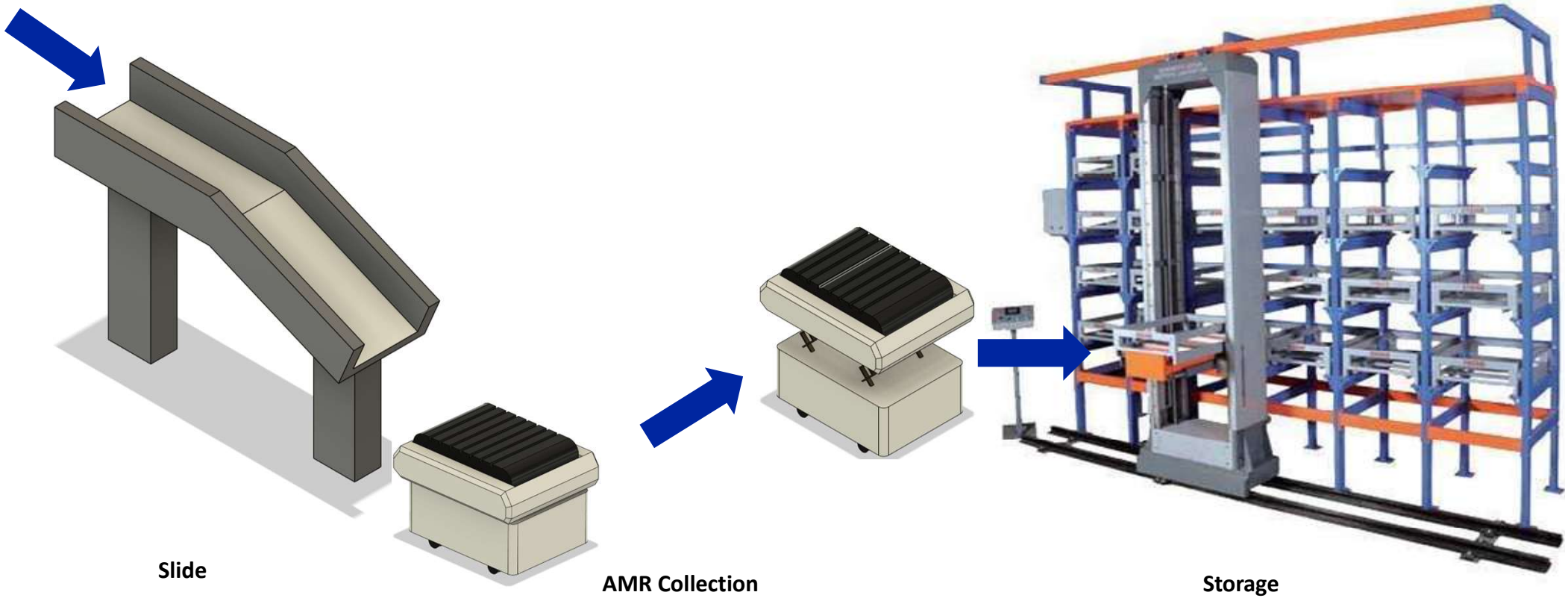
Order flexibility

No stacking

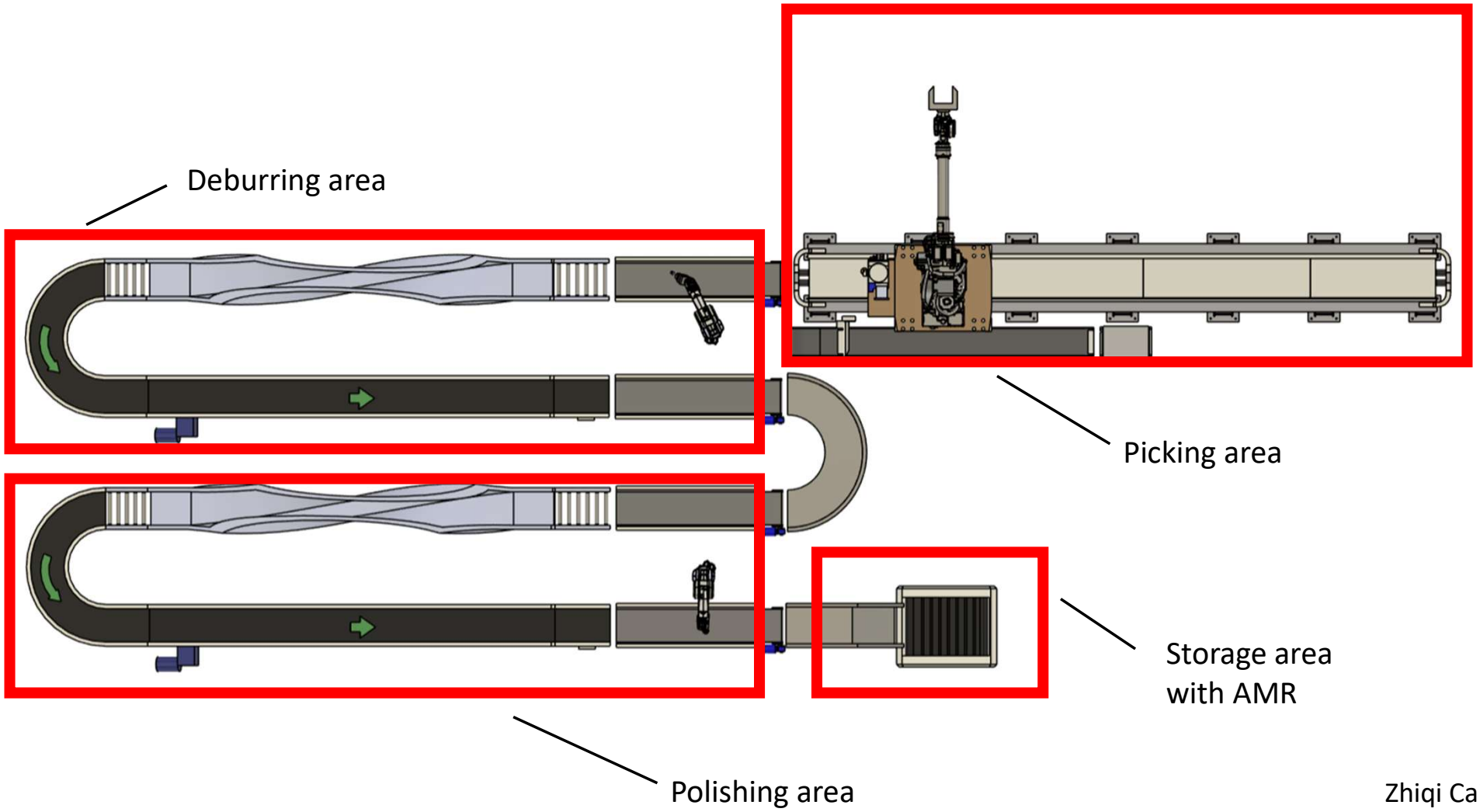
Order Uncertainty

Flexible storage is required to support both single-piece and multi-piece dispatches; stacking is not viable.

Design



Conclusion


Deburring area
Picking area
Storage area
with AMR
Polishing area